

ALETHIC MODALS AND DATIVE-INFINITIVE STRUCTURES

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I discuss the meaning of the Russian dative-infinitive construction and argue that it expresses a single kind of modality in indicative sentences, notably — alethic modality. There is no need to split this construction in the indicative mood and ascribe different categorical meanings to its varieties.

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1. Semantics-to-grammar interface

Russian is one of the few Slavic languages that preserved two productive dative patterns — dative-infinitive structures (DIS) $N_{\text{DAT}} — V_{\text{INF}}$, where the infinitive is a lexical head, and dative-predicative structures (DPS) $N_{\text{DAT}} — \text{PRED} (— V_{\text{INF}})$, where the optional infinitive is a complement of a non-verbal predicative (Zimmerling 2024). DIS sentences have a modal meaning, its definition being a matter of debate (Moore, Perlmutter 2000: 377; Fortuin 2007; Israeli 2013: 199), while DPS denote a variety of temporarily restricted situations including the situations where X must or can do something or needs something. The class of DPS predicatives contains the basic modals *nado* ‘must’ and *nužno* ‘needs’ (Laufer 2007). It is tempting to get a generalized dative sentence pattern for Russian, but all proposals to postulate the same piece of structure to DIS and DPS (Sigurðsson 2002; Mitrenina 2017) run into trouble. For the first, the dative subject of DPS is obligatory animate, while the subject of DIS is not, cf. *Gruzovikam_{DAT} zdes’ ne proexat’_{INF}* ‘Trucks can’t get through here’. For the second, the dative case is assigned structurally in Russian non-finite clauses including DIS, while in finite DPS clauses it is assigned lexically: some predicatives select dative subjects and some do not (Zimmerling 2020). For the third, DIS subjects are raised, while obligatory animate DPS subjects are syntactic and semantic arguments of the matrix clause predicatives, and there are no framework-external reasons to treat them as raised. Selectional restrictions, such as the animacy constraint, are characteristic of syntactic control, which holds for Russian (Lyutikova 2022).

The diagnostics is based on the conditions of the Russian semantics-to-grammar interface. Standard DIS are usually taken to be root clauses (Švedova 1982: 376; Tseitlin 1990: 149). One can analyze them as biclausal (Fleisher 2006; Mitrenina 2017; Zimmerling 2022), but what matters is that they lack an overt external head in syntax. Russian also licenses embedded DIS clauses with predicatives *nado*, *xorošo* ‘good’, and *lučše* ‘better’.

- (1) *V vorotax lučše_{PRED} [Inf_P stojat’_{INF} Ivanu_{DAT}, a ne Saše_{DAT}].* <embedded DIS>
 ‘It would be better <for the team> if the goalkeeper is John, not Alex.’

The example (1) conveys the meaning ‘it would be better if *p* took place’ and has narrow scope. Narrow scope signals that *Ivanu* is an infinitival argument generated in the embedded clause. The same predicatives license DPS sentences, where the dative subject is a matrix clause element, and the scope is wide, cf. (2).

- (2) *Ivanu_{DAT}, s ego rostom, lučše_{PRED} [Inf_P igrat’_{INF} vratarem].* <DPS>
 ‘It is better for John himself, tall as he is, to play the goalkeeper.’

One might object that the contrast of (1) vs (2) is due to different pragmatic interpretations of the same structure. This is the position of the linguists who claim a perfect understanding of the sentence but ignore its syntax. I start from the opposite end and assume that DPS and DIS are derived differently but some Russian sentences are two-way ambiguous between the DPS and embedded DIS readings.

2. The meaning

I argue that Russian DIS predicates express in the indicative clauses a single kind of modality — alethic modality. The four alethic modes are (logical) necessity, contingency, possibility and impossibility (Wright 1981; Mentzel 1999). I concur with Mitrenina (2017: 70) that the DIS construction covers in Modern Russian three basic alethic categories from four, except for alethic possibility. The mode depends on the verbal aspect and clausal polarity.

Table 1. The Russian DIS construction in indicative clauses

	Affirmative	Negative
Imperfective	NECESSITY <i>Tebe vyhodit’_{IMPERF}.</i> <i>Tebe exat’_{IMPERF} zavtra s utra.</i>	CONTINGENCY <i>Tebe ne vyhodit’_{IMPERF}.</i> <i>Mne ne exat’_{IMPERF} zavtra s utra.</i>

	Affirmative	Negative
Perfective	!* *Tebe sočinit' _{PERF} muzyku. *Mne uexat' _{PERF} zavtra s utra. *Tebe vyjti _{PERF} .	IMPOSSIBILITY Tebe ne uexat' _{PERF} s utra. Tebe ne vyjti _{PERF} .

In optative and counterfactual contexts, the constraint on the use of the perfective aspect in non-negative clauses is canceled.

- (3) Vykinut'_{PERF} **by**_{IRR} ej_{DAT} staryj škaf i kupit'_{PERF} novyj, tak net. <DIS>

The absence of alethic meanings in the non-veridical context (3) is expected. The gap with perfective infinitives in indicative DIS clauses is filled by DPS clauses with *nado* and other modal predicatives (Kobozeva, Laufer 1991a). The DPS construction is not polarity-sensitive and does not discriminate aspect. The sentence (4) is ambiguous between the deontic reading 'John must compose a new music piece' and the alethic reading 'It is desirable that John composes a new piece', 'the external circumstances force *p*'.

- (4) *Nado*_{PRED} sočinit'_{PERF} novuju p''esu Ivanu_{DAT}. <DPS>

There are two observations relevant for the assessment of (1) and (4). Kobozeva, Laufer (1991b) argue that non-deontic readings of *nado* are bound to contexts, where the speaker disguises *X*'s obligation as someone's need or external force. Zimmerling (2020: 52) argues that narrow scope, i.e. alethic or non-deontic reading of Russian predicatives including *nado* is overtly marked by the focus accent on the dative subject. Contrariwise, the topical accent on the preposed DPS subject of *nado* normally signals the deontic meaning, cf. the example (2).

DIS sentences do not show a correlation between communicative status and modal flavour. Moreover, they do not express other kinds of modality except for alethic. Here, however, a linguist must step down from logic towards natural language. Sentences like *Gruzovikam zdes' ne proexat'*, *Ivanu ne sdat' ekzamen* are not analytic truths, they are only *contingently true* in terms of Mentzel 1999: it is possible that in some world *w*, the trucks will get through that place, and John will pass his exam. Thus, in the linguistic perspective, alethic modalities describe *logical or physical* necessity and possibility: the real world is such that it is not in the cards for John to pass the exam. The 'in the cards' tag is used in some references to Russian DIS (Moore, Perlmutter 2000: 377; Fleisher 2006: 4), though Israeli (2013: 209) finds it inappropriate. It should be noted that Moore & Perlmutter and Fleisher do

not go far from the Russian tradition. The Academic Grammar tells that Russian infinitival clauses typically ‘convey the meaning of objective predestination’ (*ob’ektivnaja predopredelennost’*) Švedova (1982: 373), and Tseitlin (1990: 149) adds a synonym — *prednačertannost’*, while Apresjan (1995: 532) provides the paraphrase ‘the speaker is sure that the important event *p* is inevitable’. Such characteristics imply that we live in a modal world, where some or all external situations including the one described by the DIS sentence are predestined irrespective of our will, deontic obligations, or internal needs.

3. Discussion

Linguistic taxonomy of the modal meanings is controversial since linguists stick to the same terms borrowed from philosophy but use them ad libitum. Bulygina & Shmelev (1997: 218–231), who do not address DIS sentences, acknowledge the alethic vs. deontic asymmetry but show that the basic Russian modal verbs *dolžen* and *možet* are multifunctional and regularly express deontic, alethic and epistemic meanings. This echoes von Wright’s point that alethic and deontic categories behave similarly in modal logic (Wright 1981). A question then arises. If modal expressions are normally underspecified, and their ultimate interpretation depends on the context (Kratzer 1991), why should linguists recognize non-ambiguous alethic operators in Russian? I give my answer towards the end of this paper.

Kobozeva and Laufer consider *dolžen* along with *nado* and *sleduet* ‘should’: *nado* is a DPS predicative, and the impersonal verb *sleduet* selects the dative subject too. *Sleduet* is obligatory deontic, while the uses of *dolžen* and *nado* are classified into the deontic vs. non-deontic groups, the latter one labeled ‘analytic’ (Kobozeva & Laufer 1991a: 171). It hosts the meanings classified as alethic or epistemic by Bulygina & Shmelev. The affinity of alethic (circumstantial) and epistemic meanings is mentioned by Kratzer who points out that the modal base, i.e. the ideas of necessary and possible things is formed by the epistemic background (Kratzer 1991: 644–645). Deontic predicates are the dedicated part of the modal inventory since the deontic subject must be aware of the norm that makes him commit *p*. This point is clarified by Kobozeva’s definition of deontic coercion: “There are external circumstances *Q* that force *X* to commit *P*. If *X* does not commit *P*, the external force *F* that is part of *Q* will cause evil to *X*.” (Kobozeva 2000: 247).

A premise of the outlined account is that the deontic subject is animate and capable of reflection, otherwise (s)he won’t get the norm and the risks of violating it. We stated before that the DIS construction licenses inanimate subjects. Neither does it exclude event nouns in the subject position, cf. *Byt’_{INF} groze_{DAT}* ‘There will be a thunderstorm’. Thus, if the DIS construction has a general meaning in indicative

clauses, it can only be alethic, not deontic. Apresjan (1995: 533) however raises doubts whetherthetic sentences like *Byt'*_{INF} *groze*_{DAT} realized with the VS order represent the same construction as categorical SV sentences like *<Ja znaju:> Sadu*_{DAT} — *cvest'*_{INF} ‘<I know:> The garden shall bloom’. I briefly discuss the attempts to split the DIS construction below. I first state that imperative DIS clauses have the animacy constraint, which gives a ground to treat them as a related but different construction. Unless you are a poet, it makes no sense to address a garden rose and demand that it bloom and smell, but the same request is appropriate when addressed to a human being.

- (5) *Cvesti*_{INF} *Vam*_{DAT} *i paxnut'*_{INF}, *Roza Ivanovna!*
 ‘Be safe and prosperous, Roza Ivanovna!, lit. ‘Bloom and smell!’

Imperatives are deontic, and it is expected that imperative DIS clauses have the animacy constraint. This effect is absent from subjunctive DIS clauses with the particle *by*: they license inanimate dative subjects and event nouns.

- (6) *Net by*_{IRR} *ètoj*_{DAT} *mašine*_{DAT} *slomat's'a*_{INF} *popozže!*
 ‘Couldn’t this car broke down later!’
- (7) *Net by*_{IRR} *ètoj*_{DAT} *groze*_{DAT} *projt*_{INF} *poran'she!*
 ‘Couldn’t this thunderstorm broke out earlier!’

For the sake of space, I skip the discussion how the complex meanings of Russian imperative and subjunctive DIS clauses are formed but assume that they can be modeled on the basis on the meaning of Russian indicative DIS, while the reversed is false. There is no path to the alethic meaning of Russian indicative DIS clauses from any non-veridical uses of DIS.

Let us now consider the attempts to split the DIS construction or rebrand it in a way that detaches it from the logical type proposed here — alethic modality. Israeli (2013: 199, 209) objects against ‘in the cards’ cliché and claims that imperfective affirmative DIS “constitute predictions upon an omen or conjecture”, imperfective negative DIS refer to ‘condition *p*’, and perfective negative DIS convey “the Speaker’s opinion that the Subject is physically unable to perform *V*”. Thus, one gets three DIS constructions in the indicative mood. I have the impression that the author involuntarily conflates the object language with the meta-language. If one treats ‘in the cards’ not as part of the English lexicon and a tool of translating Russian DIS sentences but as conventional label referring to a single kind of modality, there is no problem with classifying all three licit combinations of aspect and polarity with alethic modes in the wake of Mitrenina 2017, see tab. 1. Israeli’s point (2013: 213) that perfective negative DIS express internal (*X*’s inability to perform *p*) but not alethic modality is difficult to assess. Probably the author adopts some version of the

conversational background analysis, which provides a path from external to internal impossibility. One can also guess that the inventory of modal types is subcategorized differently, as in Belyaeva 1990: 140, where both alethic possibility and alethic impossibility are identified with internal modality. Such discrepancies arise because the linguists pick up various semantic terms or coin them to substantiate the contrasts they find important. However, if we want to keep the descriptions of linguistic modals close to logic, it makes sense to distinguish utterances about intrinsic properties, cf. *A mug must have a handle* $\cong$ ‘the world w is such that mugs in w have handles’ from utterances about the chances of spatiotemporal events to happen, cf. *Ivanu ne sdat’ ekzamen* $\cong$ ‘the world w is such that X will not pass the exam in w ’. Events are particulars (Davidson 1970), and the force that prevents them from occurring in w can only be external. Sigurðsson, who borrows his description of the Russian DIS construction from Moore & Perlmutter, nevertheless asserts that it has a “certain deontic modality of obligation or destiny” (Sigurðsson 2002: 695, 717). This is not an accurate formulation, at best. That the modality of DIS is external does not entail that it is deontic, and ‘destiny’ is not a deontic category.

Apresjan’s point that thetic DIS sentences represent a different construction than categorical DIS sentences is backed by his diagnostics of existential vs. copular uses of the verb *byt’* ‘be’. He argues that categorical DIS sentences realized with the SV order contain the copula BE, while thetic DIS sentences show the frozen VS order and pattern with existential BE (Apresjan 1995: 532-533). This claim deserves a mention, cf. Zimmerling (2024: 13), provided that the diagnostics of the existential BE in Apresjan, Iomdin 1989 is different. Anyway, communicative status and word order do not change the type of modality in DIS. Irrespective of the fact, whether the dative subject is topicalized and fronted, cf. *Ivanu_{DAT} bylo ne sdat’_{PERF} ekzamen* $\sim$ *Ne sdat’_{PERF} Ivanu_{DAT} bylo ekzamen*, the indicative DIS construction conveys the meaning ‘external force F prevents X from doing p ’.

4. Conclusions and perspectives

There are no valid grounds for splitting the indicative DIS construction in Russian or classifying its varieties with internal and deontic modals. It invariably signals an alethic meaning, the mode of which depends on the verbal aspect and clausal polarity. The indicative DIS construction is polarity-and-aspect sensitive but lacks the animacy condition and is not sensitive to communicative status and word order. The constraint on perfective infinitives in non-negative DIS clauses disappears in the imperative DIS construction, where the animacy condition arises. This proves the switch from alethic to deontic modals in a different type of clauses. The main competitor of indicative DIS sentences is the DPS construction with modal predicatives

like *nado*. These lexical items occasionally license embedded DIS clauses. Russian modal verbs and predicatives produce both deontic and alethic sentences. The non-ambiguous character of the indicative DIS construction is probably due to the fact that its modal operator is covert or not expressed lexically. Its formalization depends on the framework, but two solutions hinted by Moore, Perlmutter (2000: 386) still seem viable: the force of DIS sentences either comes from a silent modal predicate or is a constructional feature. The latter answer is advocated in Fortuin 2007, but its assessment is beyond the reach of this paper.

Abbreviations

DAT — dative; DIS — dative-infinitive structures; DPS — dative-predicative structures; IMPERF — imperfective aspect; INF — infinitive; InfP — infinitive phrase; IRR — irrealis; PERF — perfective aspect; PRED — predicative.

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